Cryptography from Sampling Oracles and LOCC Primitives in the Bounded CRQS Model

Zijing Di, Fuyuki Kitagawa,
Ryo Nishimaki, Nick Spooner, Takashi Yamakawa

EPFL NTT Cornell University

CRS [BFM88]

The common reference string model

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public algorithm: g

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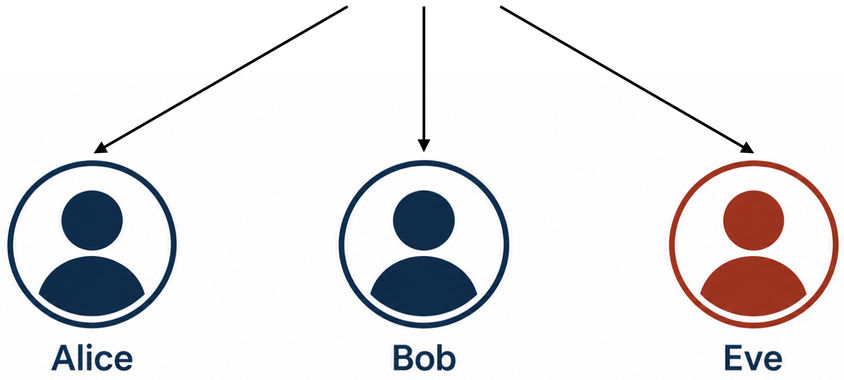
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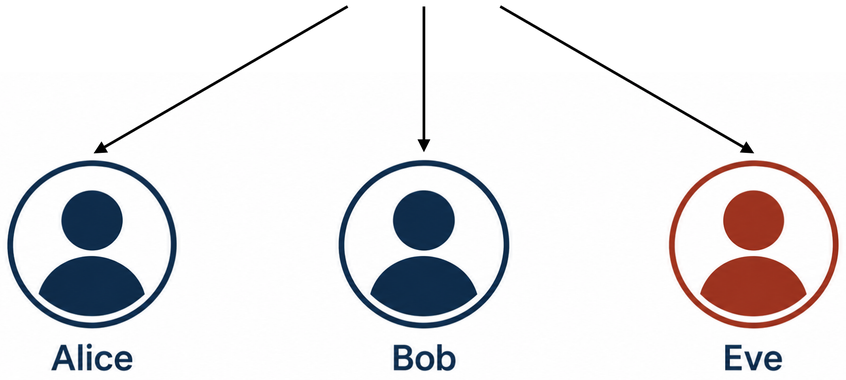
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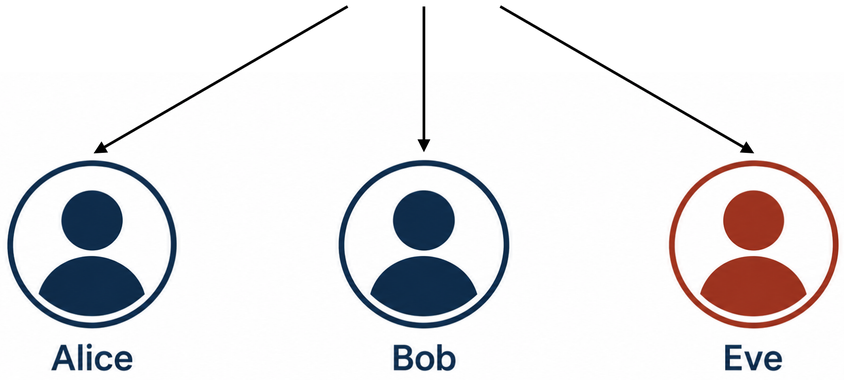
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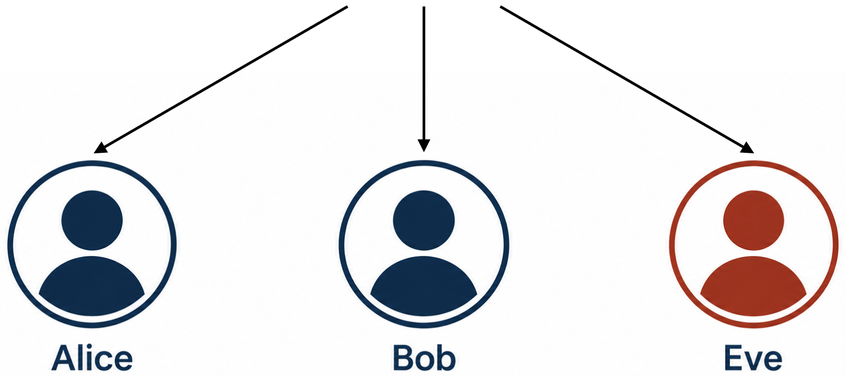
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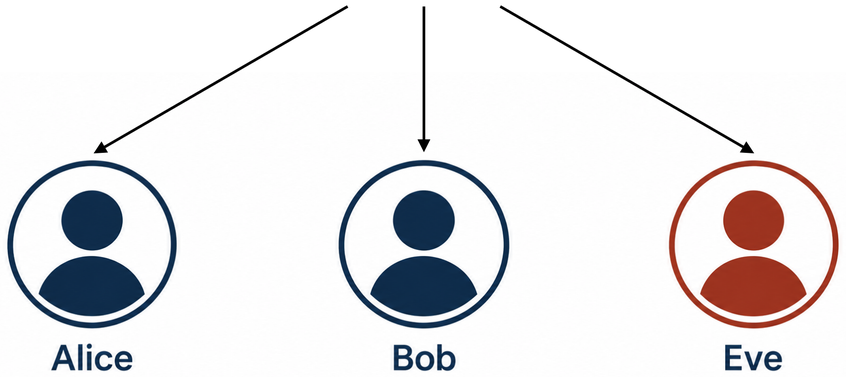
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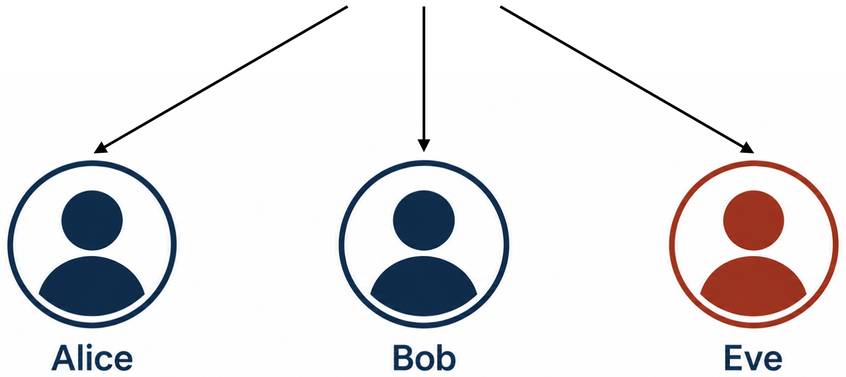
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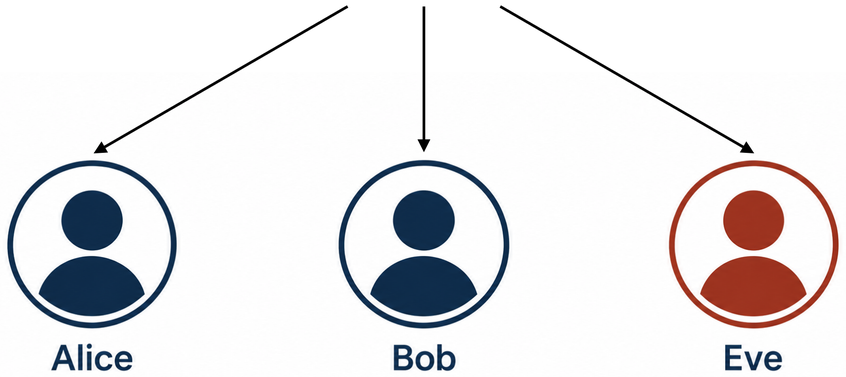
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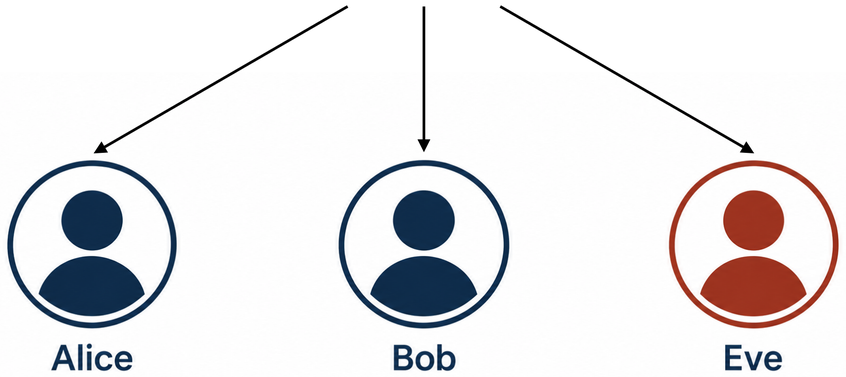
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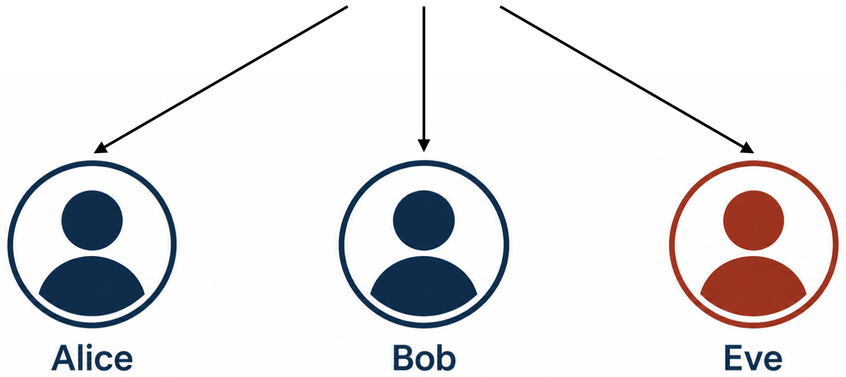
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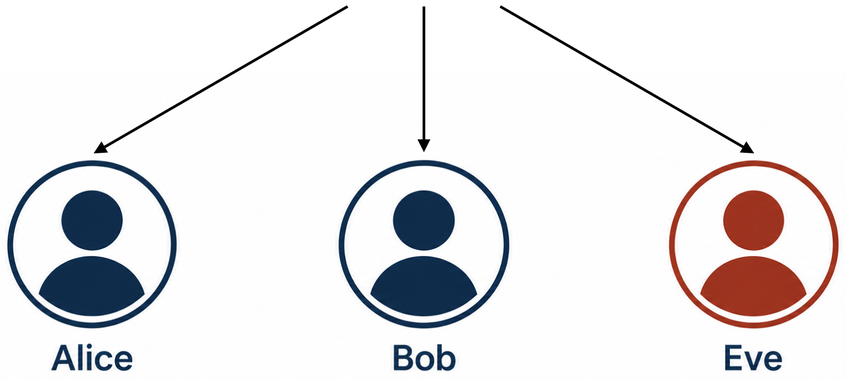
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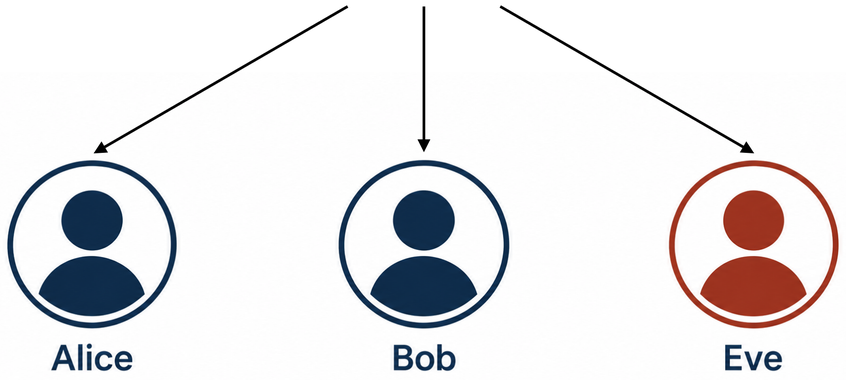
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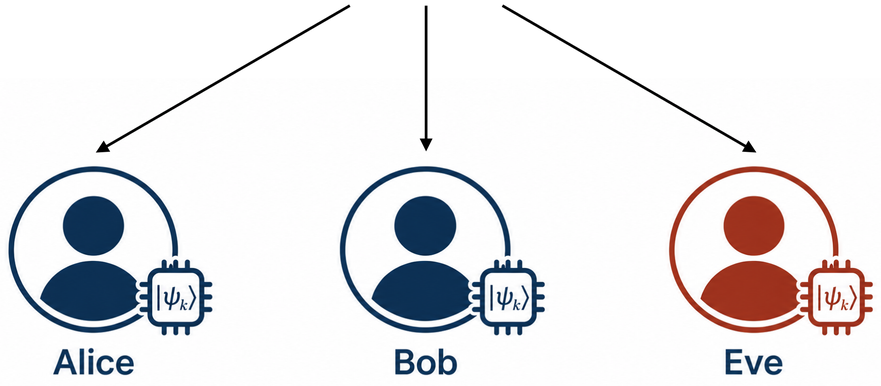
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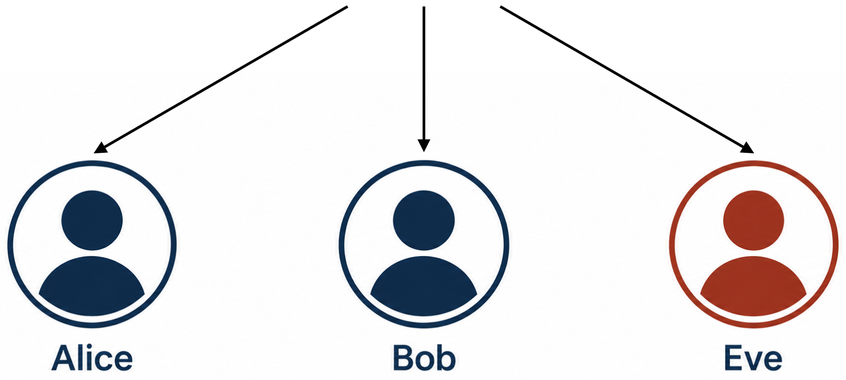
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Bounded - adversary receives *a priori* t copies
Unbounded - adversary receives any polynomial number of copies.

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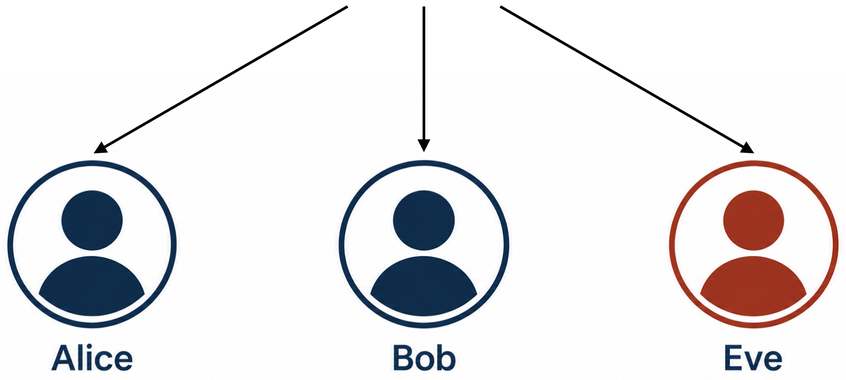
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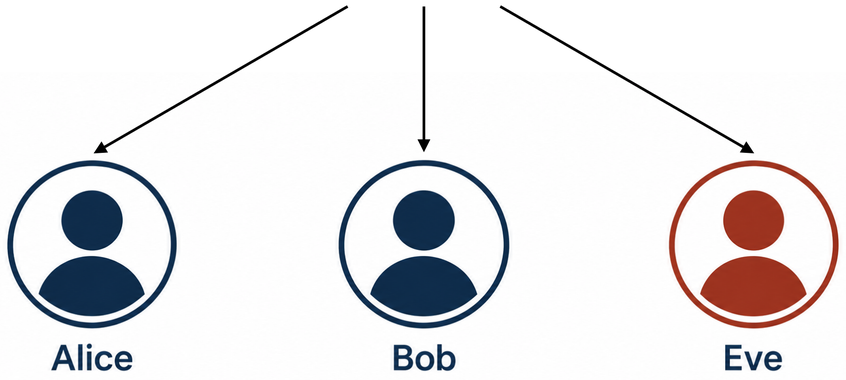
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More constructions:

Statistically-secure (hiding,binding) quantum commitments in **bounded** setting [MNY24;Qian24]

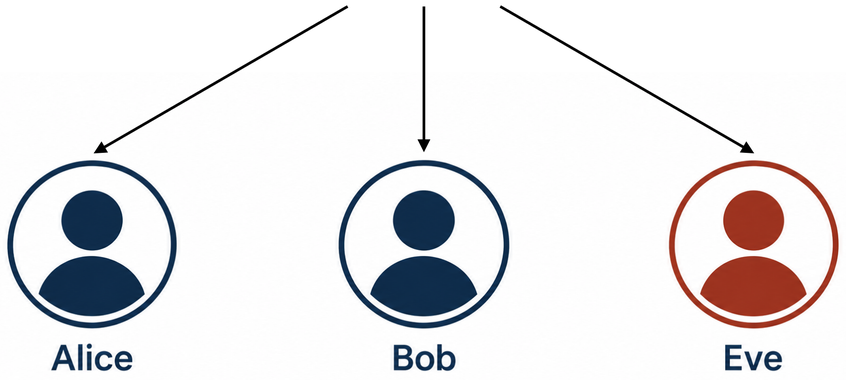
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Impossible in CRS

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Classical view

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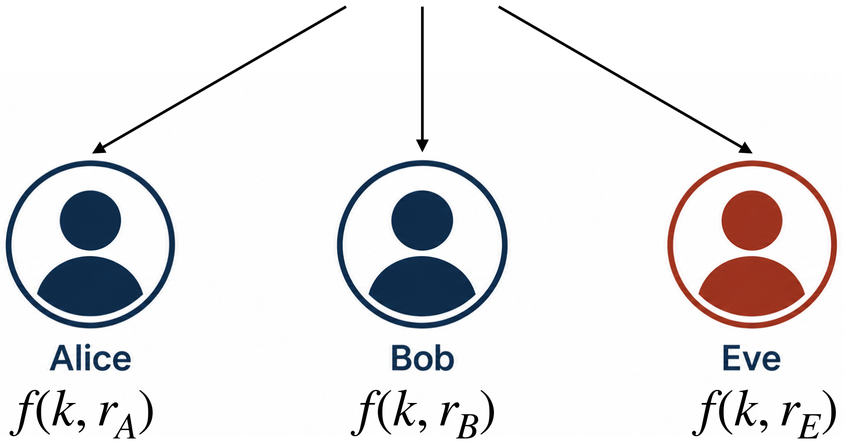
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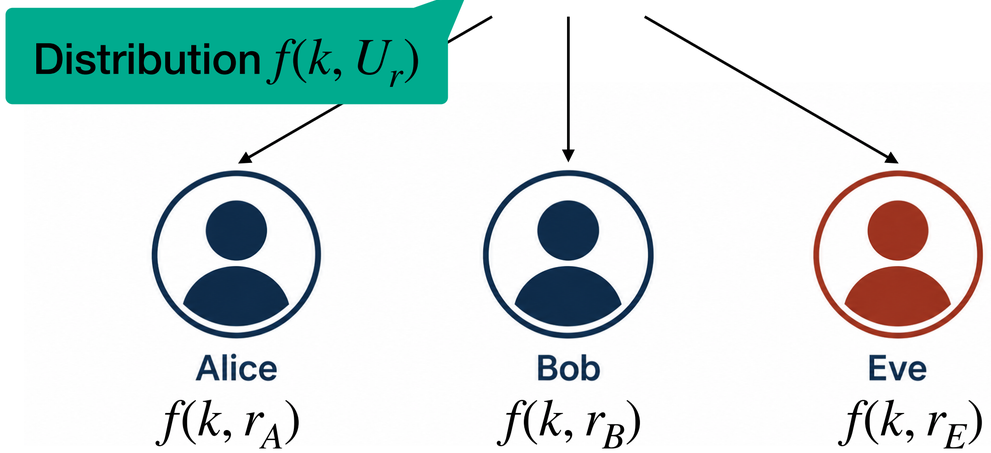
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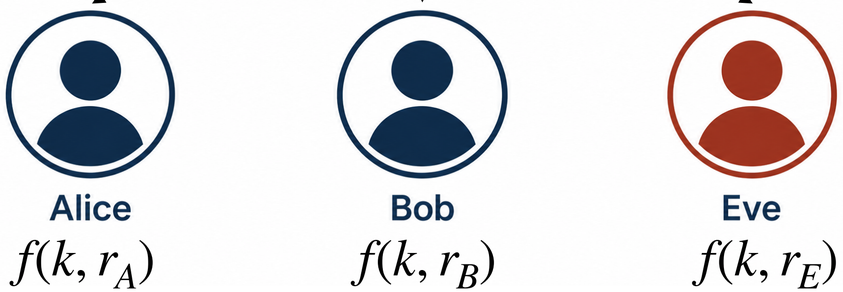
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$f(k, r_A)$ $f(k, r_B)$ $f(k, r_E)$

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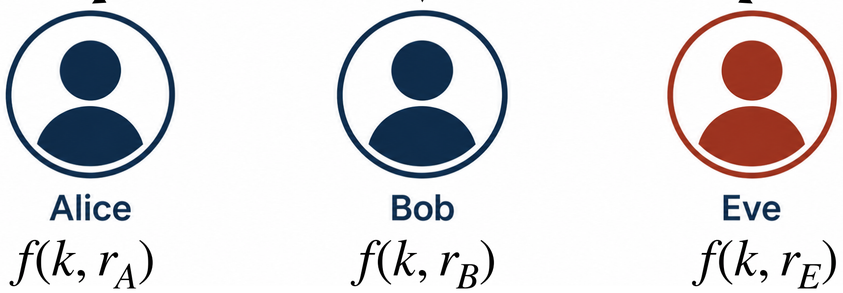
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$f(k, r_A)$ $f(k, r_B)$ $f(k, r_E)$

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$$k \leftarrow \{0,1\}^{\mathsf{k}}$$

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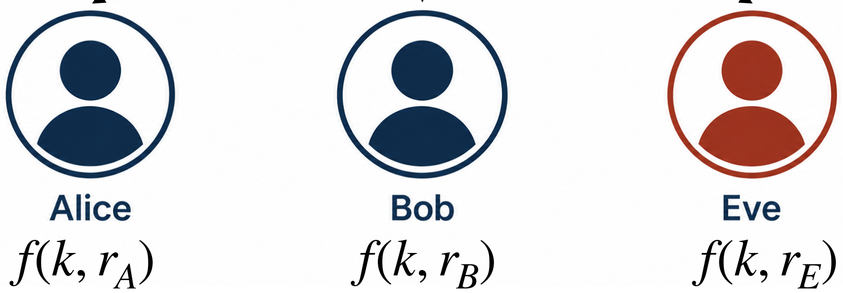
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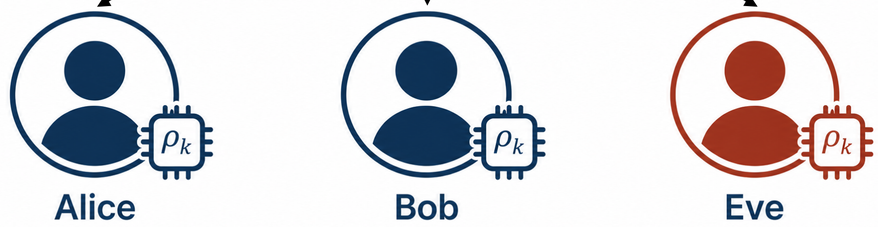


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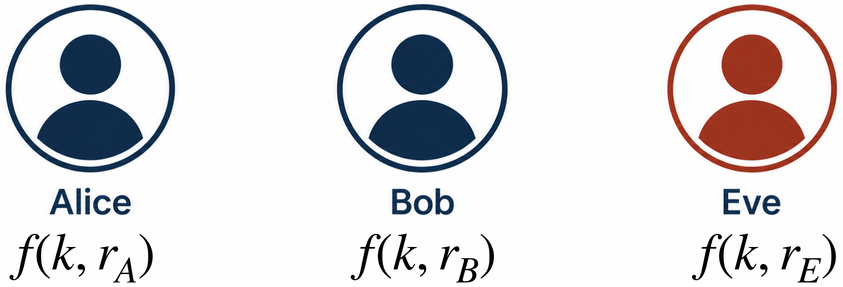
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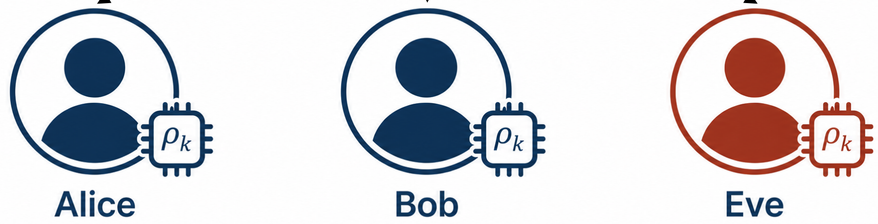


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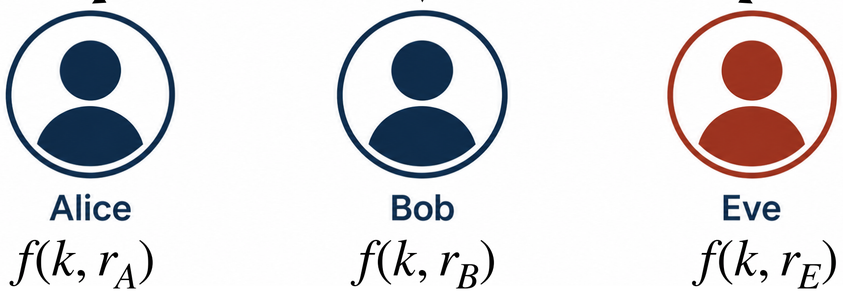
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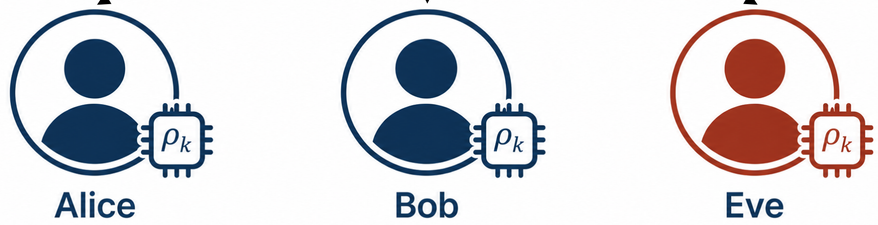


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CRD seems in between of CRS and CRQS

🤔

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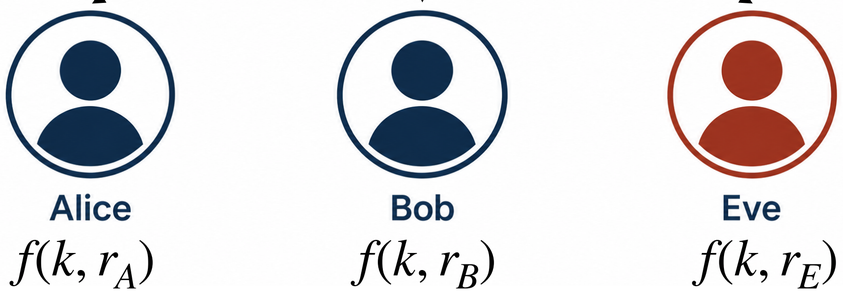
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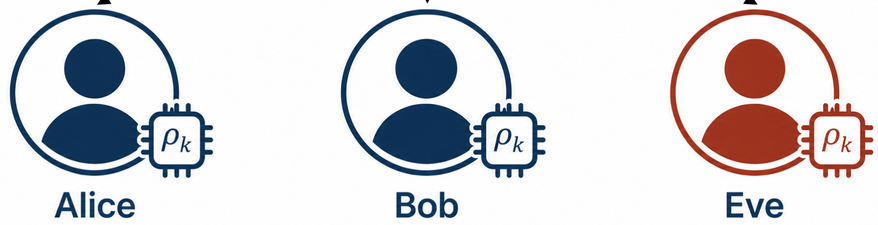


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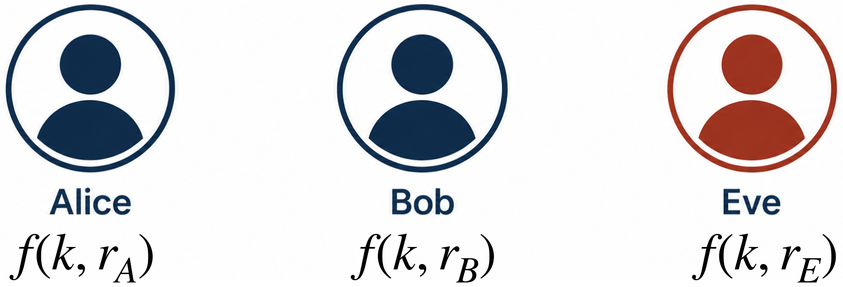
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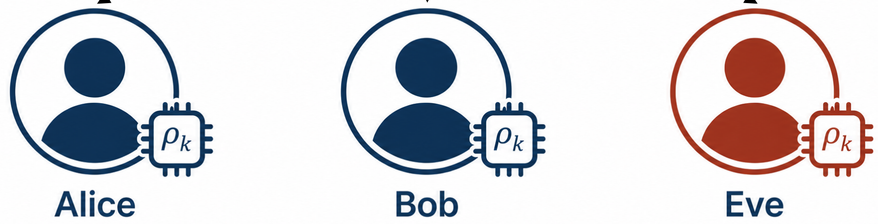
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Classical mixed state

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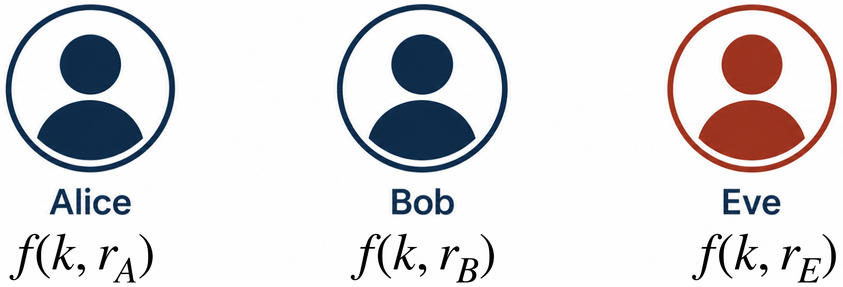
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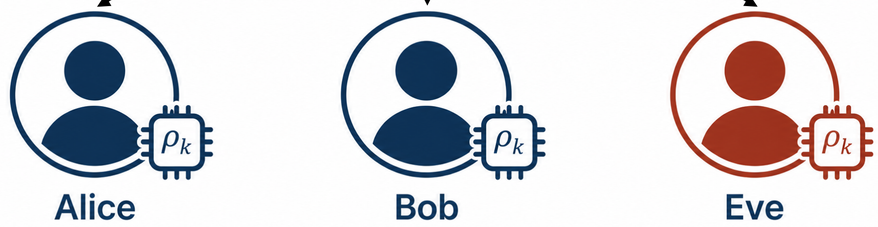
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Theorem 1. Any protocol in the bCRD model can be unconditionally transformed into an equivalent protocol in the bCRQS model, preserving classical communication. The same holds in the unbounded setting assuming one-way functions (OWFs).

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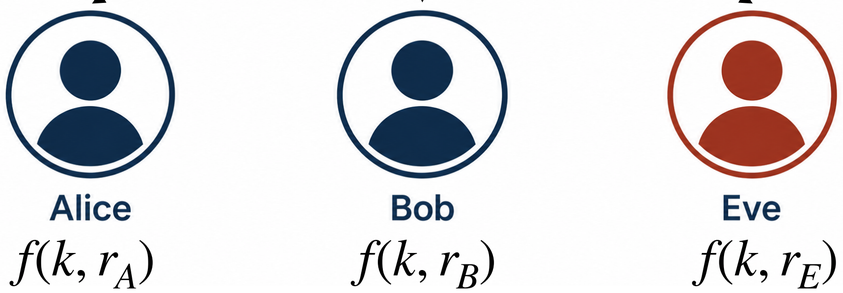
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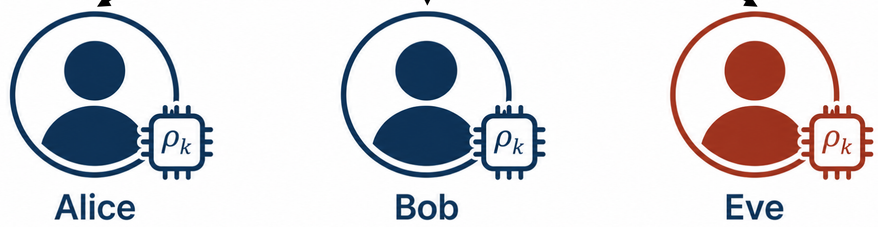
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Theorem 1. Any protocol in the bCRD model can be unconditionally transformed into an equivalent protocol in the bCRQS model, preserving classical communication. The same holds in the unbounded setting assuming one-way functions (OWFs).

b(CRD, CRQS): bounded, with a priori sample bound
ub(CRD, CRQS): unbounded, any poly number of samples

Our model: CRD

The common reference distribution model

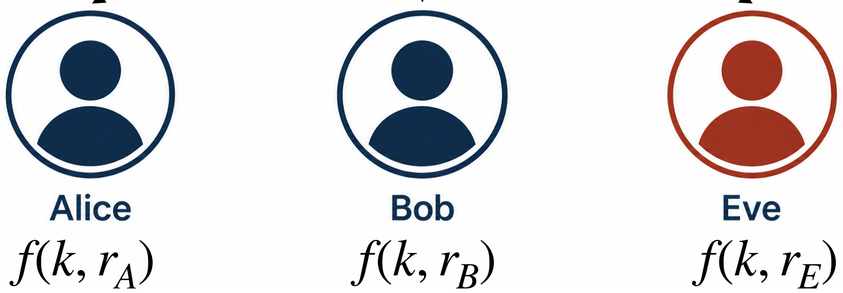
Classical view

public algorithm: f

$k \leftarrow \{0,1\}^k$

$r_A, r_B, r_E \leftarrow \{0,1\}^r$

Distribution $f(k, U_r)$

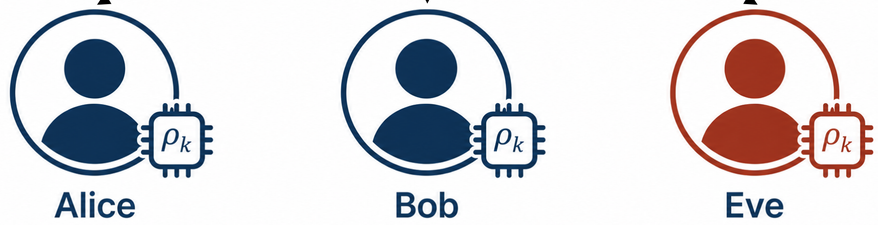


Quantum view

Classical mixed state

$k \leftarrow \{0,1\}^k$

$\rho_k = \mathbb{E}_r[\,|f(k,r)\rangle\langle f(k,r)|\,]$



CRD seems in between of CRS and CRQS

Not obvious!

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Why this model?

CRQS is great! But quantum resources is expensive…
Can we build classical primitives?



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Yes!



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Yes!



Theorem 2: Constructions

CRQS is great! But quantum resources is expensive…
Can we build classical primitives?



Yes!



Theorem 2: Constructions

Classical primitives with unconditional statistical security in bCRD

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Non-interactive commitments

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NIZKs

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Theorem 2: Constructions

Classical primitives with unconditional statistical security in bCRD

- Non-interactive commitments
- NIZKs
- One-time signatures

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Theorem 2: Constructions

Classical primitives with unconditional statistical security in bCRD

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- Key agreements

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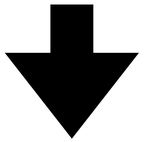


Yes!



Theorem 2: Constructions

Theorem 1 + Theorem 2



Classical primitives with unconditional statistical security in bCRD

- Non-interactive commitments
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Theorem 2: Constructions

Classical primitives with unconditional statistical security in bCRD

- Non-interactive commitments
- NIZKs
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Theorem 1 + Theorem 2

Corollary 1. Statistically-secure non-interactive commitments, one-time signatures, NIZKs, and key agreements, all with classical communication, exist unconditionally in the bCRQS model.

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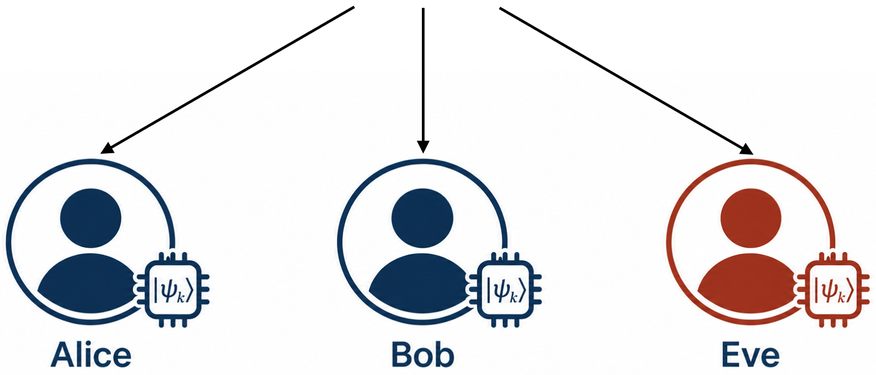
No known constructions with classical communication in CRQS before.

CRQS

public quantum algorithm: Q

$$k \leftarrow \{0,1\}^{\mathsf{k}}$$

$$|\psi_k\rangle \leftarrow Q(k)$$

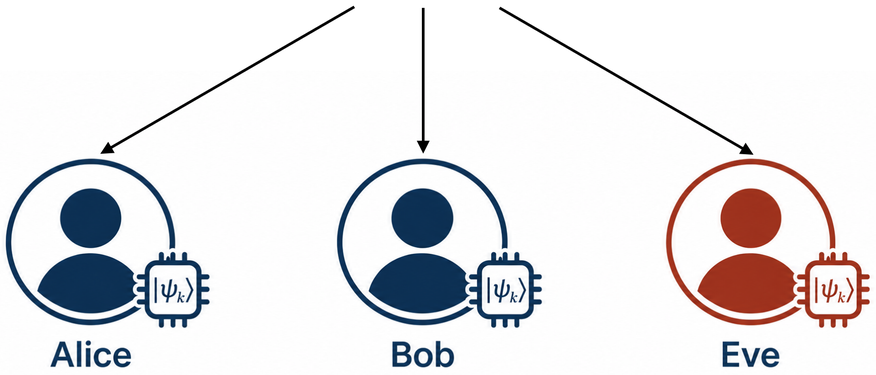


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CHRS [AGL24;CCS24]

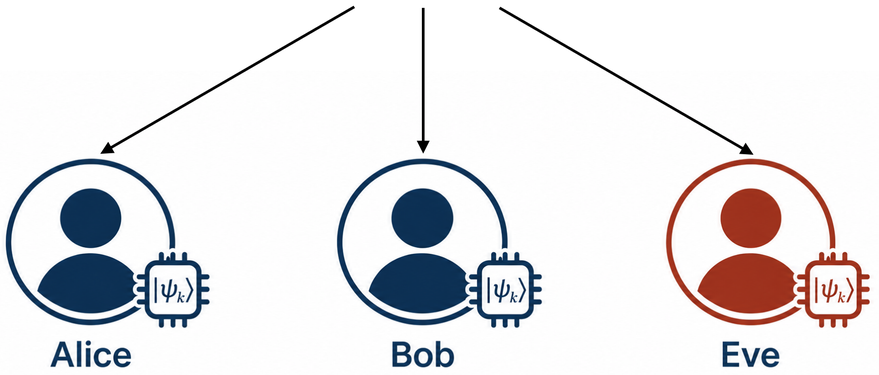
The common Haar random state model

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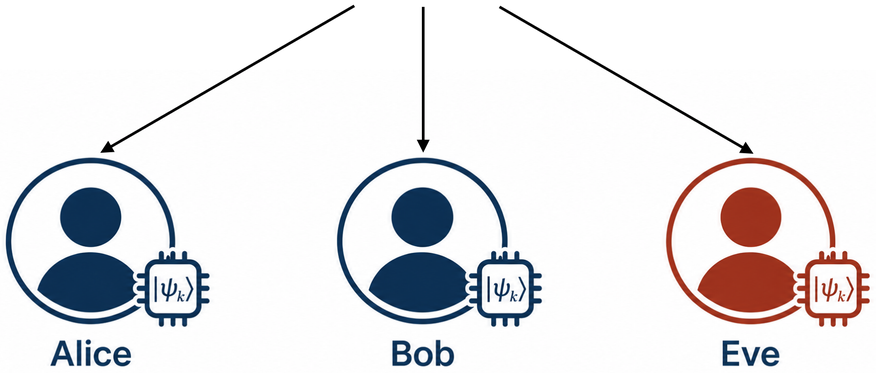
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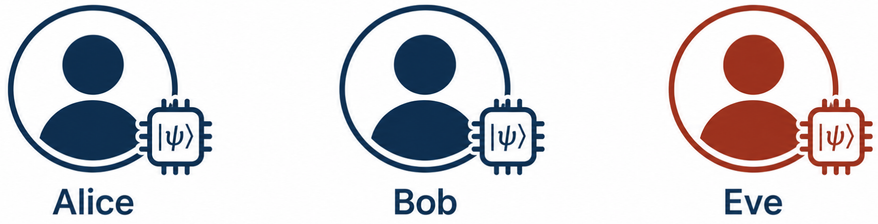
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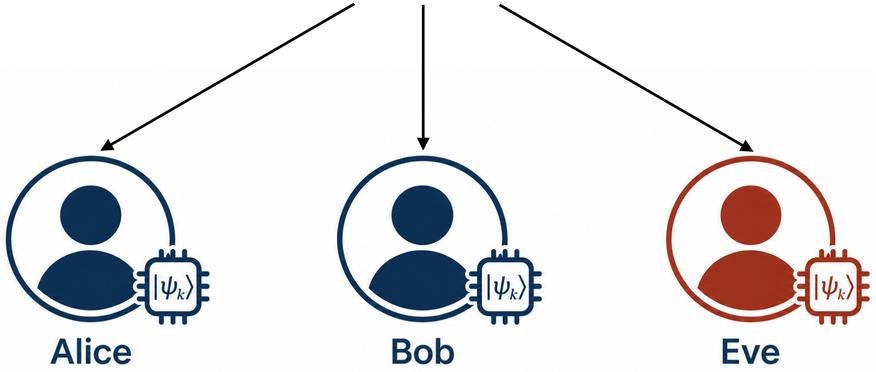


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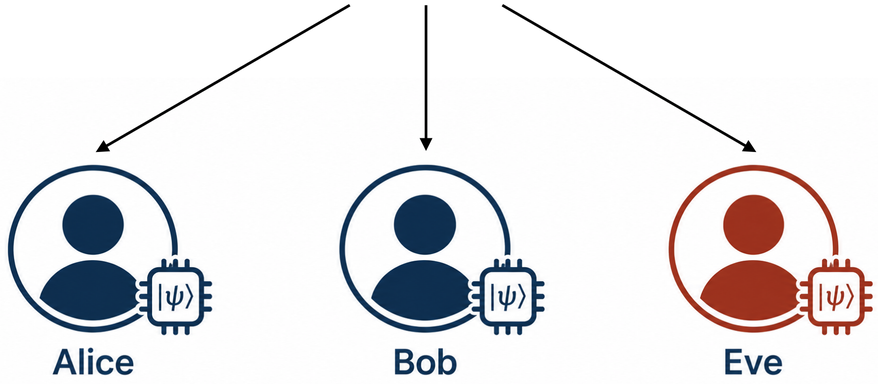


CHRS [AGL24;CCS24]

The common Haar random state model

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Not efficient

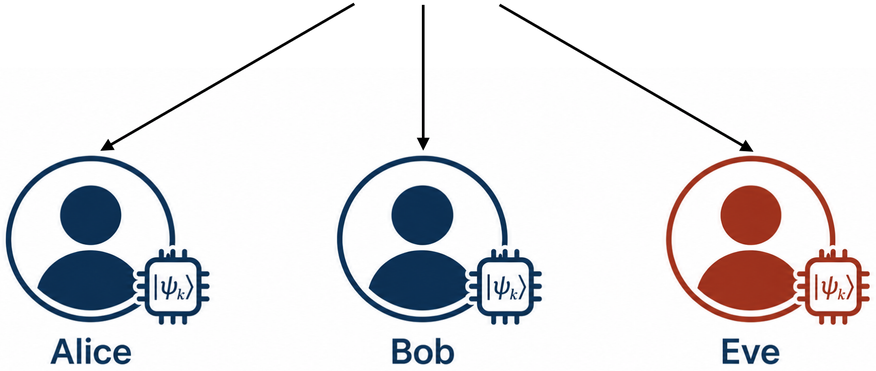


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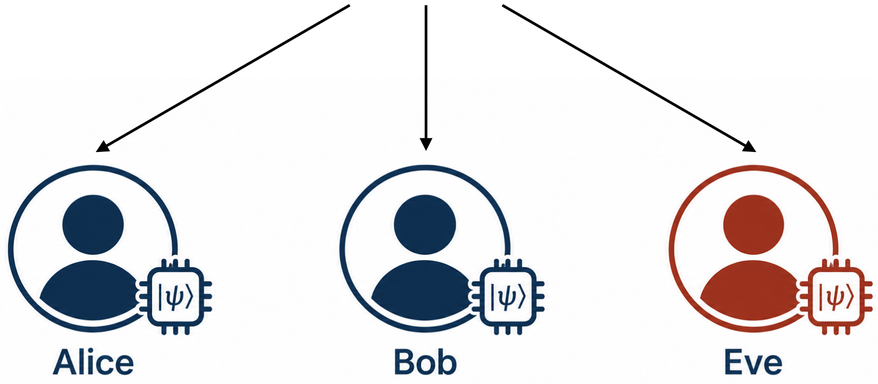


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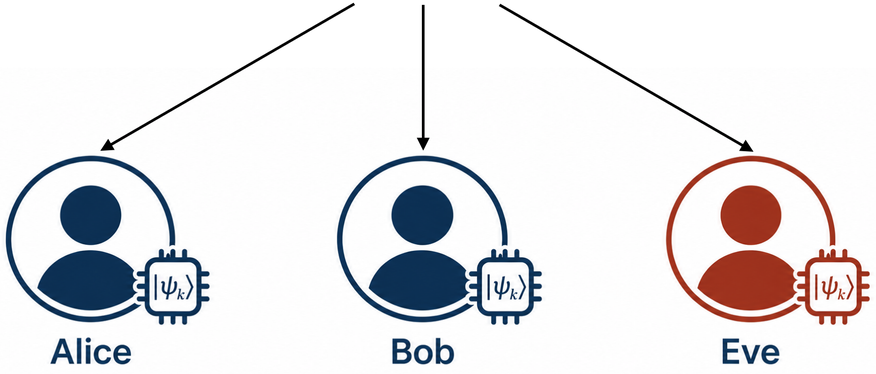
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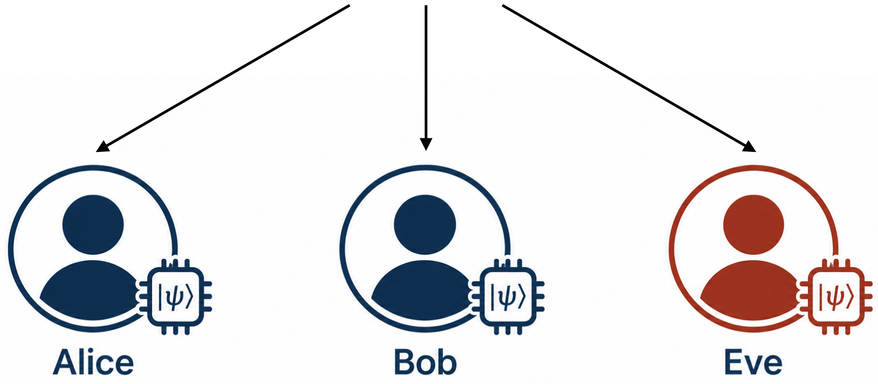


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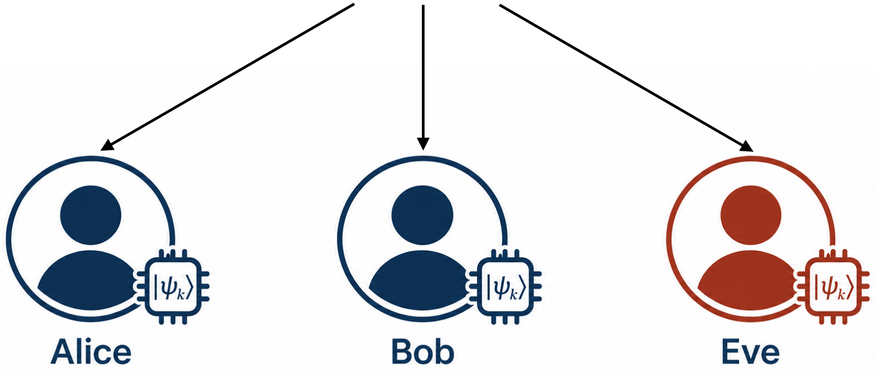
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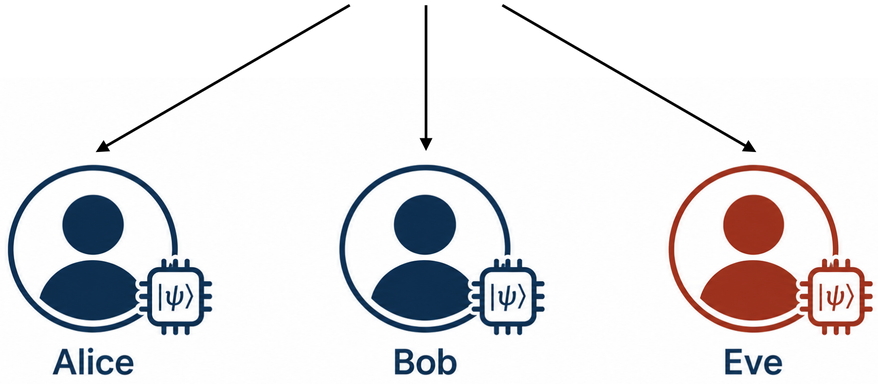


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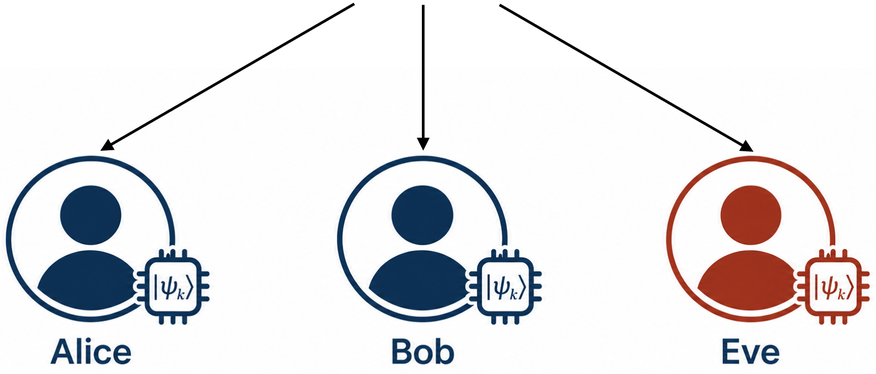
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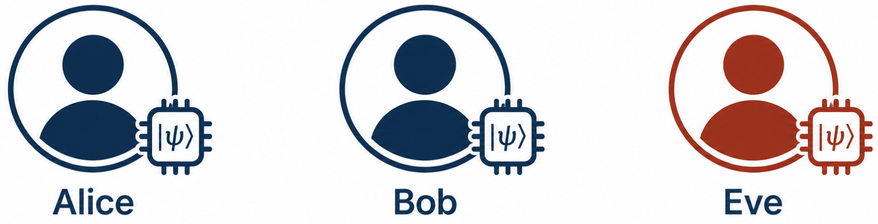


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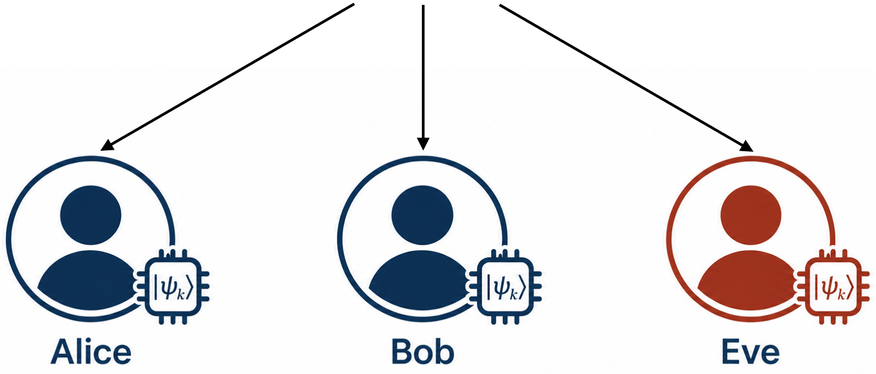
→ **bCRQS and bCHRS separation**

CRQS

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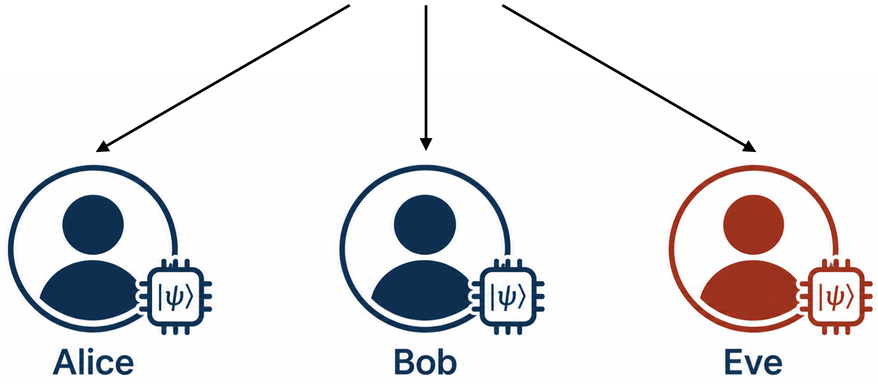


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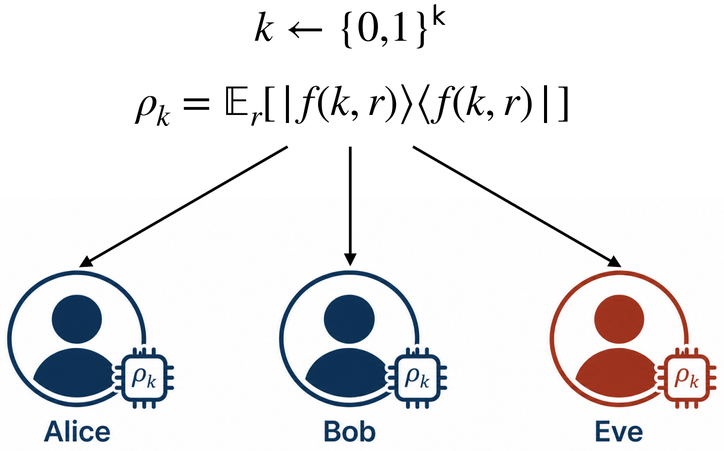
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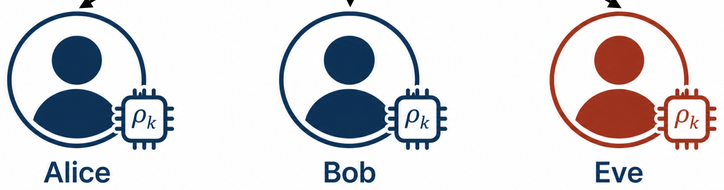


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We want a pure state

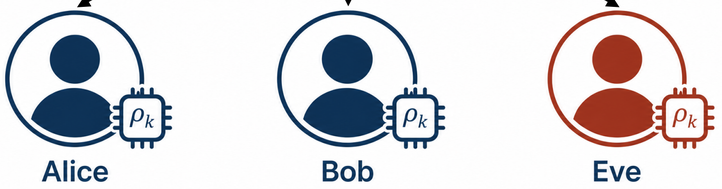


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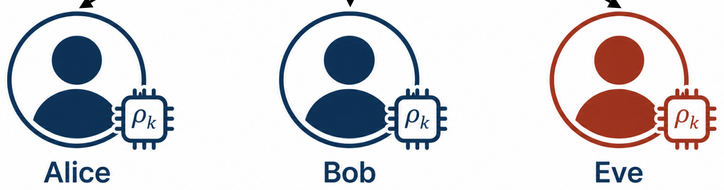


Purify!

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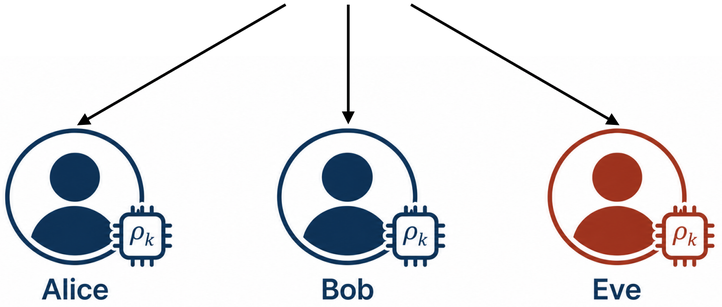
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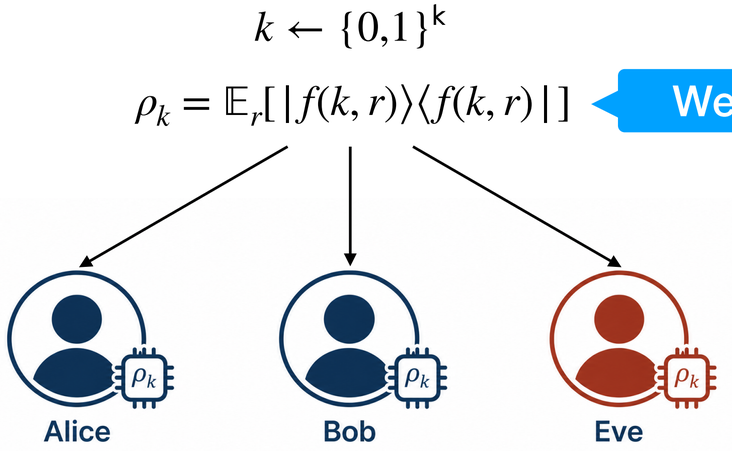


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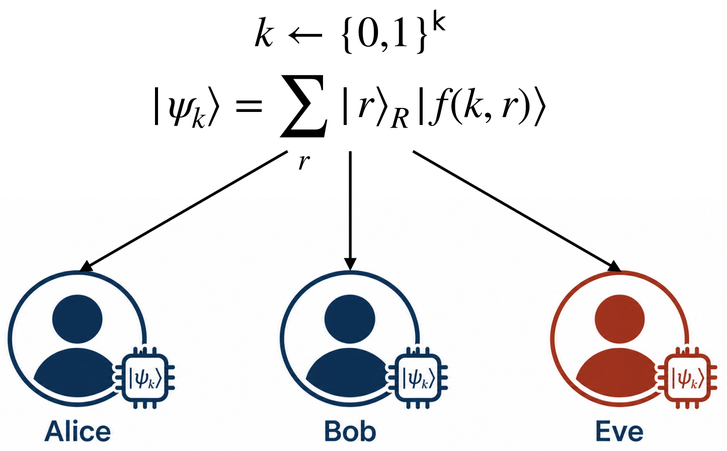
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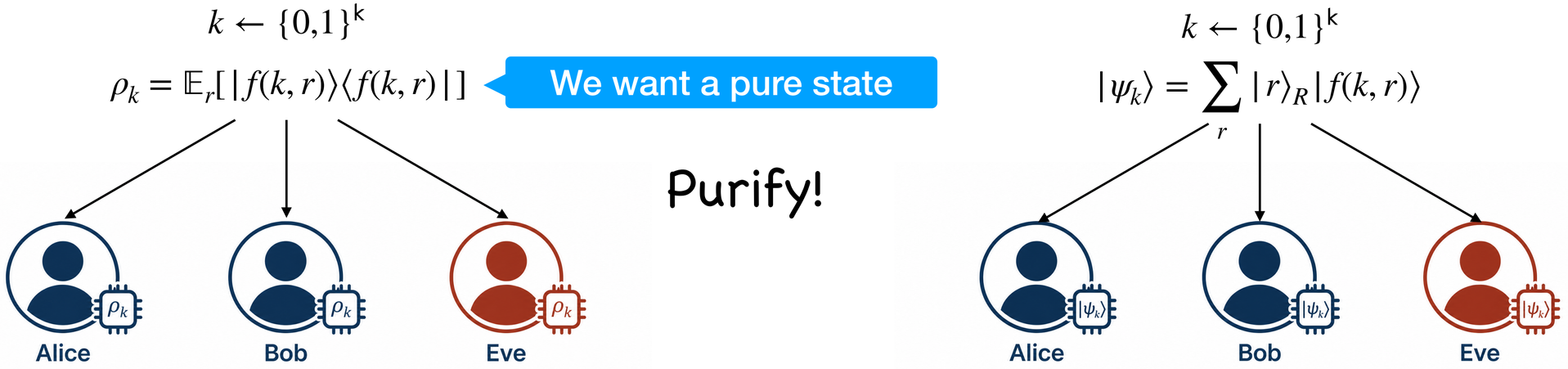


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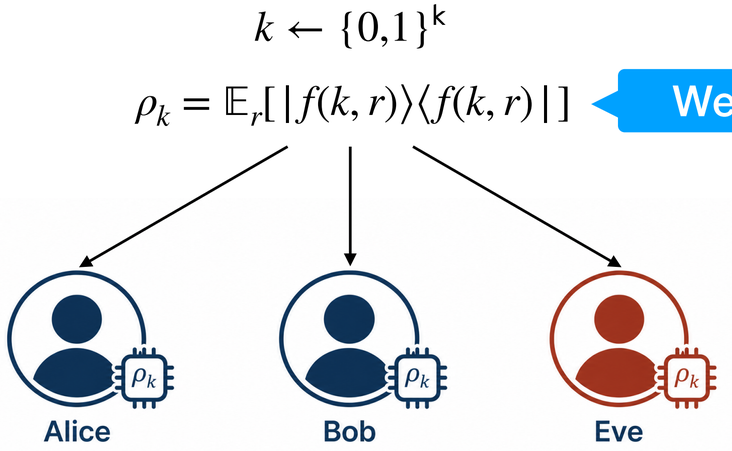


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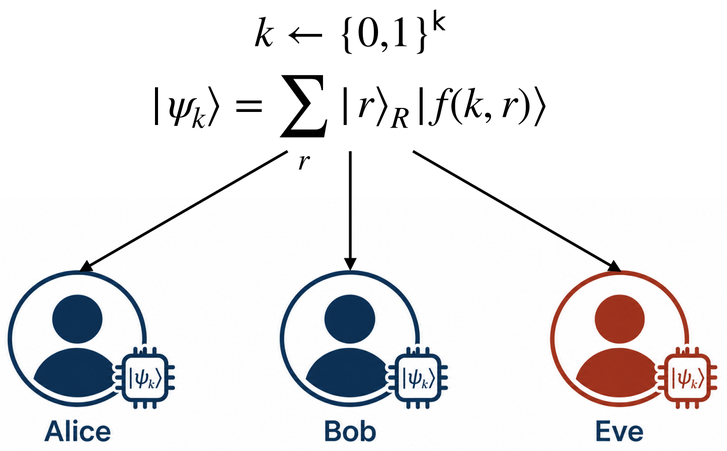
The security of our construction relies on r remaining secret.

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We want a pure state

Purify!



The security of our construction relies on r remaining secret.



Adversary gets the purifying register!

Scramble the secret to hide it!

Scramble the secret to hide it!

CRD

$$k \leftarrow \{0,1\}^{\mathsf{k}}$$

$$r_1, \ldots, r_q \leftarrow \{0,1\}^{\mathsf{r}}$$

$$s_1, \ldots, s_q = CRD(k, r_1), \ldots, CRD(k, r_q)$$

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CRQS

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$$F \leftarrow \text{Functions}$$

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q copy view

Scramble the secret to hide it!

CRD

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q copy view

$$|\{(x_1, s_1), \ldots, (x_q, s_q)\}\rangle$$

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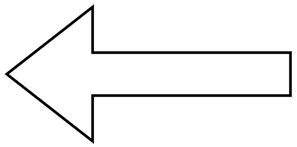
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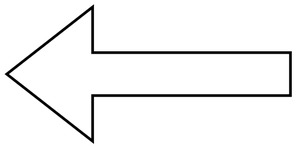
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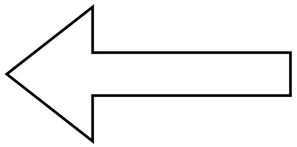
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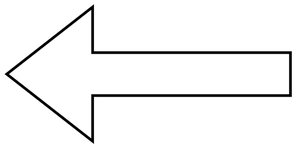
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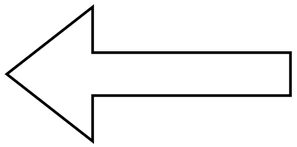
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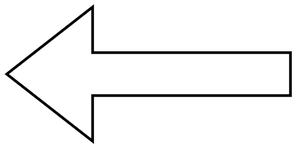
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Theorem 1 🥳

Theorem 2: Constructions

Classical primitives with unconditional statistical security in bCRD

Non-interactive commitments

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Classical primitives with unconditional statistical security in bCRD

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CRD key: random subspaces $A_0, A_1 \leq \mathbb{F}_2^m$ of dimension $m/2$.

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Statistical hiding

The full subspace covers every prefix

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$m/2$ columns



$\ell \ll m/2$ rows

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Statistical hiding

The full subspace covers every prefix

$m/2$ columns

Full row rank (w.h.p.)



$\ell \ll m/2$ rows

Theorem 2: Constructions

Classical primitives with unconditional statistical security in bCRD

Non-interactive commitments

CRD key: random subspaces $A_0, A_1 \leq \mathbb{F}_2^m$ of dimension $m/2$. **Samples:** $a_b \leftarrow A_b, \hat{a}_b \leftarrow A_b^{\perp}, \forall b \in \{0,1\}$

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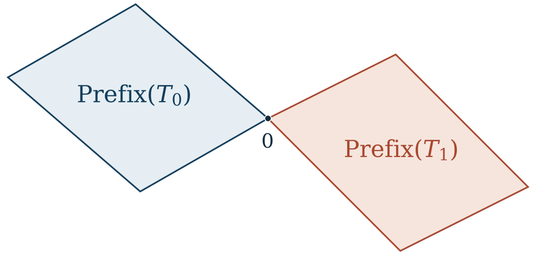
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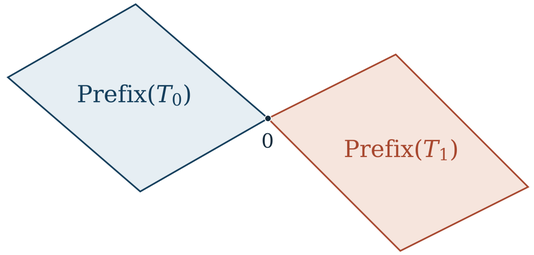
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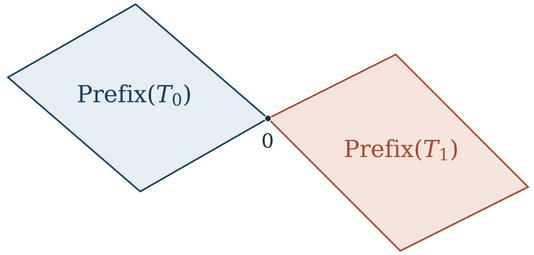
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Fun! You should also try!

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- Primitives in bCRQS: sample size and communication complexity depends only on λ.

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- (computationally-secure) bit commitments exist in bCRD with commucation complexity less than t;

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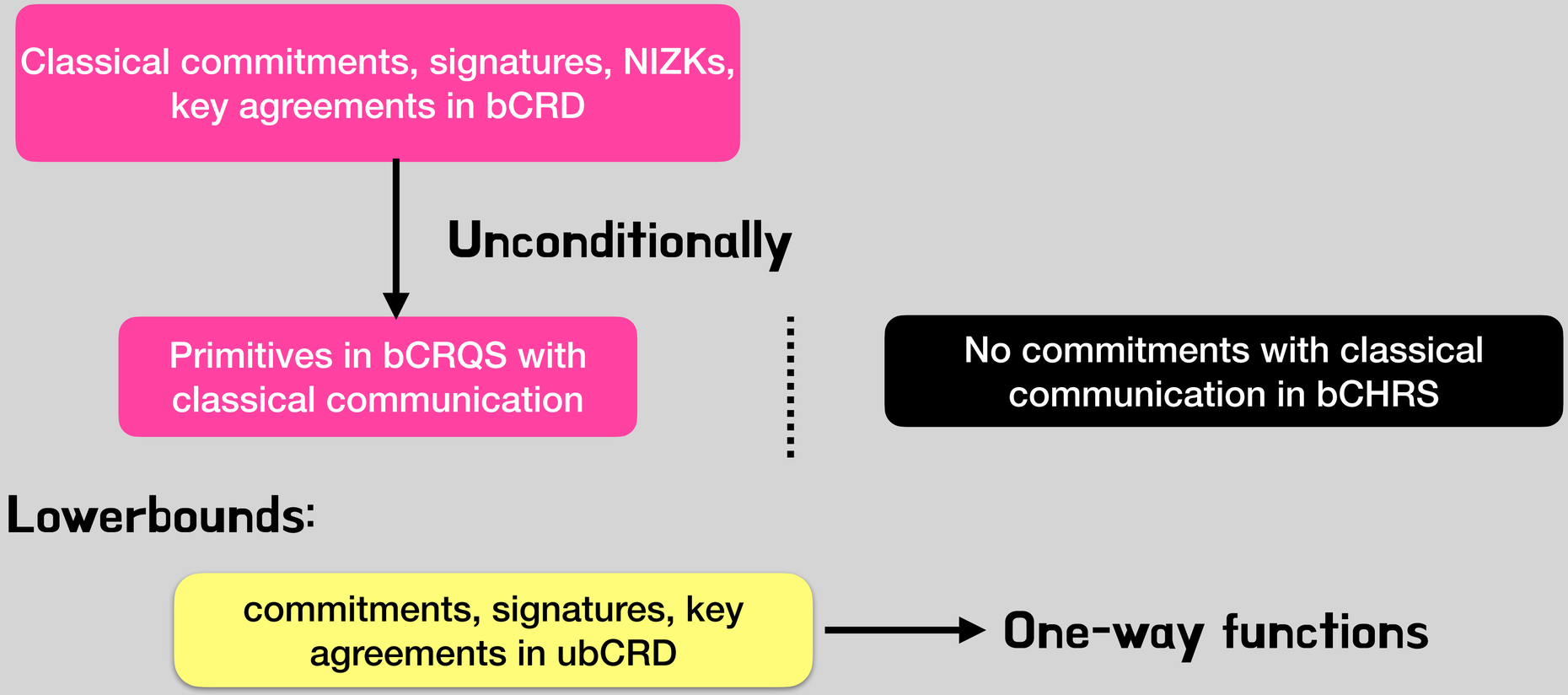
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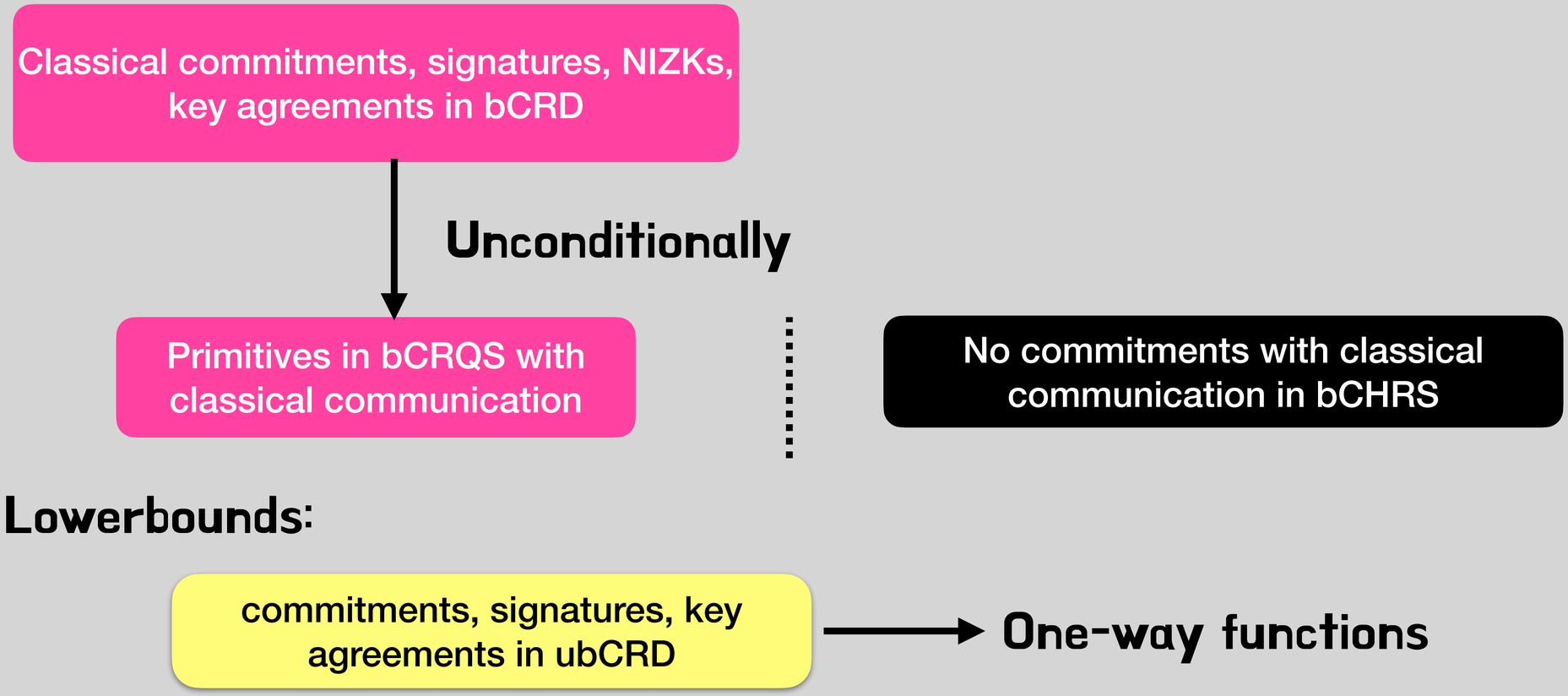
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Takeaways


Classical commitments, signatures, NIZKs, key agreements in bCRD
Unconditionally
Primitives in bCRQS with classical communication
No commitments with classical communication in bCHRS
Lowerbounds:
commitments, signatures, key agreements in ubCRD
One-way functions

Takeaways



Thank you!